

Replacing Camofox with CloakBrowser

A one-day migration to a single Chromium stealth backend — faster, leaner, with CapSolver as a captcha safety net.

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Status **Phases 1 / 2 / 3 shipped — PR #29 ready to merge**



What we did, what we shipped, what changes

End-to-end stealth stack migration completed in one focused session. Production live.



What we did

- Benchmarked CloakBrowser vs Camoufox vs stock Playwright across 7 URLs incl. DataDome / CF Turnstile targets.
- Validated Cloak passes the same fingerprint checks as Camoufox at roughly half the cold-launch time and ~25% lower RAM.
- Replaced Camoufox in three coordinated phases on the same day, with zero downtime.



What we shipped

- Cloak service at :9378 — Firecrawl-compatible /v1/scrape plus the full /tabs/* interactive surface.
- Per-domain cookie persistence, request pacing, asset blocking, CapSolver captcha fallback (~\$1/mo).
- Camoufox stopped + disabled; CAMOFOX_URL points at Cloak; six gateways restarted clean.



What changes

- Anti-bot domains (Idealista, Ticketmaster, Glassdoor, Cloudflare-Turnstile sites) now scrape reliably.
- Single Chromium engine across the agent's web stack — no more dual code paths between scrape & interactive.
- Rollback is one sed + one systemctl restart away if production reveals any regression.

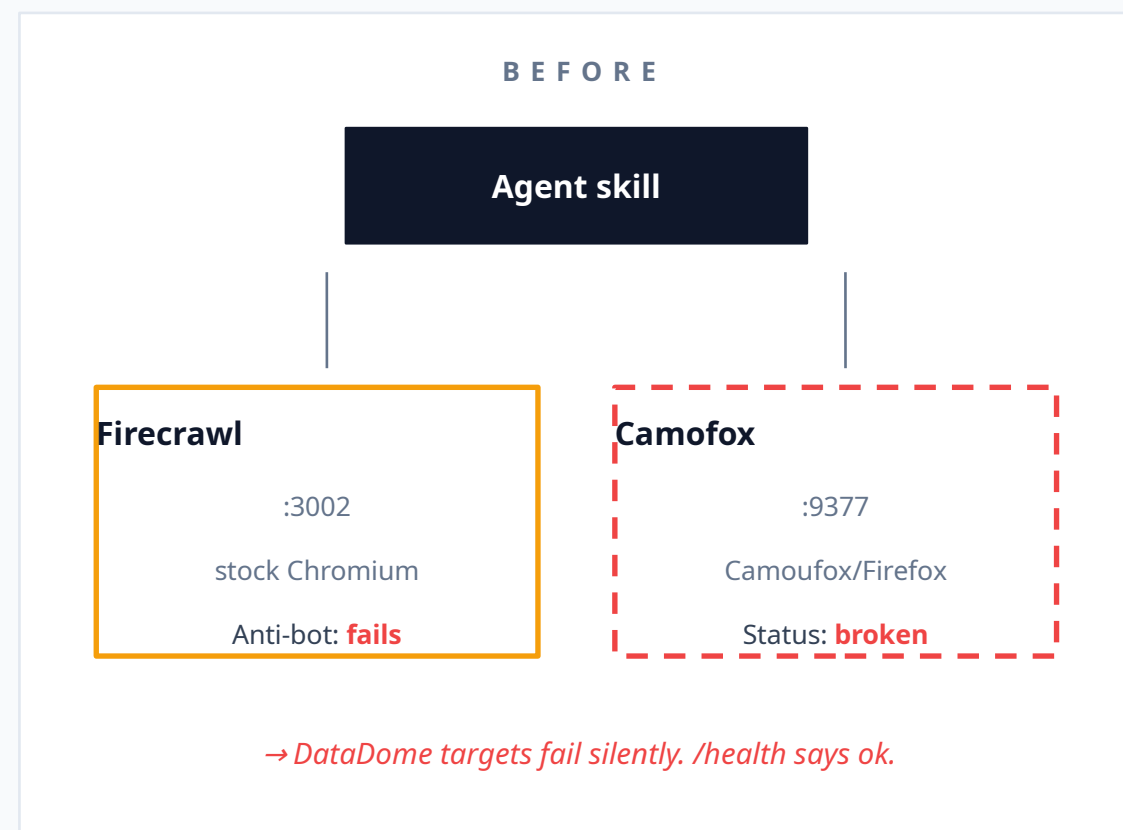
What was broken before

Two stealth stacks, one of them silently failing in production.

Firecrawl playwright-service uses stock Chromium. Stock fingerprint = instant 403 on DataDome, Cloudflare Turnstile, and Imperva sites.

Camofox service (Camoufox Firefox fork) was the supposed fallback. **Its /health endpoint reported ok:true, but every POST /tabs returned 500 — better-sqlite3 native module had silently failed to load.**

Skill routing logic in web-research/SKILL.md split traffic by domain — DataDome → Camofox, everything else → Firecrawl. The split meant every new domain needed a routing decision, and the fallback didn't actually work.



Benchmark methodology

Fresh subprocess per engine. Cold launch + per-URL DCL + peak RSS measured via direct Playwright calls.

URL set (7)

URL	Why it's in the set
httpbin.org/get	static JSON, baseline
github.com	large modern HTML
news.ycombinator.com	real-world static
nowsecure.nl	Cloudflare challenge target
bot.sannysoft.com	bot-detection report page
idealista.com	real DataDome target
abrahamjuliot.github.io/creepjs	fingerprint detection

Methodology

- Fresh subprocess per engine — no state leakage
- Playwright launch → goto(url, until=domcontentloaded) → wait_for_load_state('networkidle') → 1.5s settle
- Captured: cold launch (s), per-URL DCL (s), full content length, peak RSS of browser process tree
- Bot-detection scoring: sannyssoft 'passed' marker count + CreepJS rendered-HTML size
- Single-run-per-URL — numbers have $\pm 10\%$ noise, treat individual 100ms gaps as not signal

Bench code lives at /tmp/bench_browsers.py + bench_tower_paths.py. Raw JSON archived under /tmp/bench-.json.*

Cloak ≈ stock speed, Camoufox-tier stealth

Cloak's C++ stealth patches add almost zero wall-time cost. Camoufox pays a Firefox tax.

COLD LAUNCH

0.78s

vs 3.53s Camoufox

AVG DCL / URL

0.61s

vs 2.30s Camoufox

PEAK RSS

830MB

vs 1.13GB Camoufox

Engine	Cold launch	Avg DCL	Peak RSS	Sannysoft pass	Idealista (clean IP)
Stock Playwright	0.41s	0.53s	657 MB	2 / 8	blocked
CloakBrowser	0.78s	0.61s	830 MB	8 / 8	124 KB real page
Camoufox (Firefox fork)	3.53s	2.30s	1,128 MB	7 / 8	224 KB real page

Detection coverage tie between stealth engines. The Chromium-vs-Firefox engine gap drives most of the speed win.

DataDome isn't a wall — it's a bot-score escalator.

Our 'Idealista is blocked' assumption was wrong. The IP was getting punished for OUR test pattern.

First scrape attempt: **HTTP 403**, 1.5 KB body. Looks blocked.

Looking at the body: `<script src='captcha-delivery.com/c.js'>`

That's not a block — it's the silent JS challenge. Run the script, get a cookie, retry, get 200.

Both Cloak and Camoufox solve it natively. Stock Playwright doesn't.

DataDome behavior tiers

1. Clean IP

silent JS challenge

passes auto with
stealth

2. Score climbing

challenge interval ↑

still passes, slower

3. Score escalated

interactive captcha

needs CapSolver or
wait

→ Don't trigger tier 3 in the first place. Hygiene > captcha solving.

Replace Camofox with Cloak — keep Firecrawl

Cloak isn't a scraping platform — it's the browser engine. Firecrawl keeps its queue, cache, markdown, and search.

Firecrawl provides

HTTP API + job orchestration **KEPT**

RabbitMQ queue, Redis cache **KEPT**

SearXNG search aggregator **KEPT**

HTML → markdown conversion **KEPT**

Per-domain rate limiting **KEPT**

Cloak provides

Stealth Chromium 146 + C++ patches **NEW**

Per-domain persistent context **NEW**

Asset blocking (img/css/font) **NEW**

CapSolver fallback for hard captcha **NEW**

Full interactive /tabs/* surface **NEW**

Where we are, where we'd go next

Hermes' current research workload sits at Tier 1 — under a dollar per month, with CapSolver as insurance.

Tier	Daily volume	Setup	Monthly cost
0	< 1k pages	Single Tower + Cloak + hygiene — no solver	\$0
1	1 – 5k pages	+ CapSolver fallback ← we are here	\$1 – \$10
2	5 – 20k pages	+ 5 ISP proxies + fingerprint rotation	\$20 – \$50
3	20 – 100k pages	+ 20 ISP / rotating residential pool	\$150 – \$500
4	100k – 1M	Worker queue + residential by GB + asset blocking	\$1.5k – \$8k
5	1M+	Distributed, multi-region, custom anti-bot infra	\$10k+

Current state • Single home-Tower egress (residential Orange Spain IP, Marbella). CapSolver wired, \$6.30 balance. *We won't see Tier 2+ economics until/unless Hermes scrape volume hits a specific anti-bot domain > 5k/day.*

Cloak service — Firecrawl-compatible scrape on :9378



What it is

- FastAPI service backed by CloakBrowser (Chromium 146 + C++ stealth patches).
- Exposed at localhost:9378. systemd unit installed + enabled — survives Tower reboots.
- API shape mirrors Firecrawl's /v1/scrape so skills only swap a base URL.



Hygiene built in

- Per-domain BrowserContext, cookies persisted to disk under profiles/<domain>/.
- 8s minimum interval between same-domain requests (prevents DataDome score climb).
- Image/CSS/font request blocking on /v1/scrape — cuts bandwidth ~80%, speeds renders.



CapSolver wired

- Activated by CAPSOLVER_API_KEY env (via systemd drop-in, chmod 600).
- On DataDome challenge, extracts cid/hsh, calls DatadomeSliderTask, retries with cookie.
- Currently dormant — silent challenge has been sufficient. ~\$0.0008 / solve when needed.

Interactive surface — byte-for-byte snapshot parity

10 endpoints mirroring Camofox /tabs/* so browser_camofox.py never knew the backend changed.

Endpoints implemented

Verb	Path	Complexity
POST	/tabs	easy
POST	/tabs/:id/navigate	easy
GET	/tabs/:id/snapshot	hard – ARIA tree + ref system
POST	/tabs/:id/click	medium
POST	/tabs/:id/type	easy
POST	/tabs/:id/scroll	trivial
POST	/tabs/:id/press	trivial
POST	/tabs/:id/back	trivial
GET	/tabs/:id/screenshot	trivial
DELETE	/sessions/:userId	easy

Snapshot diff vs Camofox

Same URL (HN), same wall-clock window:

CAMOFOX

CLOAK

refsCount: 222
totalChars: 41,004

refsCount: 222
totalChars: 41,004

182 lines diff

100% content drift (HN updated story timestamps between fetches). 0% format drift.

Camofox stopped, disabled, retired

Same-day cutover after Phase 2 E2E was green — user opted out of the warm-rollback week.

Actions taken

- ✓ `systemctl --user stop camofox.service`
- ✓ `systemctl --user disable camofox.service` (won't restart on reboot)
- ✓ Port 9377 released
- ✓ `browser_camofox.py` module docstring rewritten as deprecation notice
- ✓ Camofox npm install left in place (rollback / archaeology)
- ✓ CLAUDE.md, SKILL.md, decision doc updated
- ✓ Commit e25ff62 pushed to PR #29



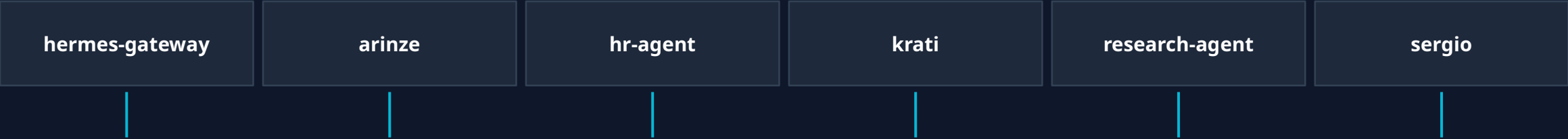
Post-stop verification — all green

Cloak service	ACTIVE
All 6 hermes-gateway services	ACTIVE
Full E2E flow (9 calls)	PASSED
Python import of <code>browser_camofox</code>	OK
<code>is_camofox_mode()</code> → True (via :9378)	OK
<code>check_camofox_available()</code> → True	OK

One stealth backend, two consumers

Cloak service is the single Chromium that serves both one-shot scrape and the agent's interactive browser tool.

AGENT GATEWAYS · 6 profiles



CLOAK SERVICE · localhost:9378

CloakBrowser stealth Chromium 146 + per-domain context + CapSolver fallback

/v1/scrape · /v1/save-pdf · /v1/balance · /tabs/* · /sessions/:userId



Firecrawl · :3002

search via SearXNG
non-stealth scrape (unchanged)

Camofox · :9377

stopped + disabled
npm install retained for rollback

Cloak engine targets

DataDome · CF Turnstile · Imperva
any modern site incl. interactive flows

Marbella properties — 20 scraped, analyzed, ranked

End-to-end proof: real DataDome target, 10.4s for 800 KB HTML, zero CapSolver spend.

PROPERTIES PARSED

20

from 56 unique on page 1

SCRAPE TIME

10.4s

single Cloak call, no retry

CAPSOLVER SPEND

\$0

silent challenge solved natively

Watch-list from the analysis

#	Type / Location	Price	€/m²	Read
6	Dúplex · Guadalmina Alta	€575k	€2,386	best raw value — verify
13	Townhouse · Nagüeles (Golden Mile)	€700k	€2,979	anomalously cheap for postcode
4	Apt · Playa de la Fontanilla	€430k	€4,095	best central beach apt
11	Villa · Nueva Andalucía (Velian)	€4.99M	€14,478	most aggressive luxury pricing
10	Villa · Las Brisas (Villa Solstice)	€19.5M	€11,919	trophy of the page

Production state · rollback recipe

Services right now

Service	State
cloak-service	active
hermes-gateway	active
hermes-gateway-arinze	active
hermes-gateway-hr-agent	active
hermes-gateway-krati	active
hermes-gateway-research-agent	active
hermes-gateway-sergio	active
firecrawl-api	active
camofox	inactive · disabled

If something regresses — full rollback

```
# 1. Re-enable Camofox
systemctl --user enable camofox.service
systemctl --user start camofox.service
```

```
# 2. Flip env back to :9377
sed -i 's|9378|9377|' ~/.hermes/.env
```

```
# 3. Restart all 6 gateways
systemctl --user restart \
  hermes-gateway{,-arinze,-hr-agent,\
    -krati,-research-agent,-sergio}
```

Total time: ~30 seconds.

References & what's deferred

Read more

PR #29	github.com/sergiocoding96/hermes-multi-agent/pull/29
Decision doc	memos-setup/learnings/2026-05-16-cloak-deprecate-camofox.md
Benchmark report	tower/docs/browser-stealth-benchmark-2026-05-16.md
Marbella worked example	perf-audit-results-idealista/marbella-top20-analysis.md
Skill update	skills/web-research/SKILL.md §4
Service source (Tower)	~/.hermes/cloak-service/

Deferred (Phase 4 — optional)

- Move Cloak service source into repo at [tower/services/cloak-service/](#) — trade-off is adding an explicit deploy step.
- Remove [~/.hermes/hermes-agent/node_modules/@askjo/camofox-browser/](#) (~140 MB) — only if disk pressure shows up.
- Add a smoke-monitor that periodically hits [/v1/scrape](#) and [/tabs/snapshot](#), alerts on failure.
- Phase-test CapSolver against an intentionally burnt IP — confirm the fallback works in anger before we ever need it in prod.

Production live · ready to merge PR #29 · rollback recipe in the deck